



x

1

7

x

10

0

1

0

1







PARADOXES

ESPRESSO







BEAR

Figure 1 consists of two horizontal bar charts. The top chart shows the percentage of respondents for different levels of agreement with the statement 'The government should do more to protect the environment'. The bottom chart shows the percentage of respondents for different levels of agreement with the statement 'The government should do more to protect the environment'.

Level of Agreement	Percentage of Respondents
Strongly agree	10%
Agree	40%
Disagree	30%
Strongly disagree	20%

Level of Agreement	Percentage of Respondents
Strongly agree	10%
Agree	40%
Disagree	30%
Strongly disagree	20%

A pixelated, grayscale image of the number 4, rendered in a blocky, digital font style. The number is composed of various shades of gray and black pixels, giving it a low-resolution, retro appearance. It is centered on a white background.

A pixelated, grayscale version of the Twitter bird logo. The bird is shown in profile, facing right. It has a rounded body, a pointed beak, and a single wing with a distinct curve. The image is composed of large, visible square pixels in various shades of gray, giving it a low-resolution, digital-art appearance. The background is white.



THE FIRST TWO



100%







2023-10-21

$$= \underbrace{\frac{p_t^2}{2 \log(2)} \pi \lambda \left(\frac{h \varphi_{\text{RX}}}{\sin^2(\epsilon)} \right)^2}_{k \triangleq} \exp \left\{ - \tau^2 \underbrace{v_{\text{sat}}^2 \sin^4(\epsilon) / (h^2 \varphi_{\text{RX}}^2)}_{D \triangleq} \log(2) / 2 \right\},$$



























Fig 1(a): True $P(t)$, GPR estimate & forecast, and 100-sample MA & forecast

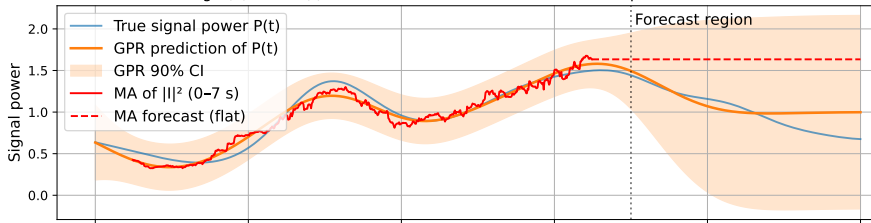
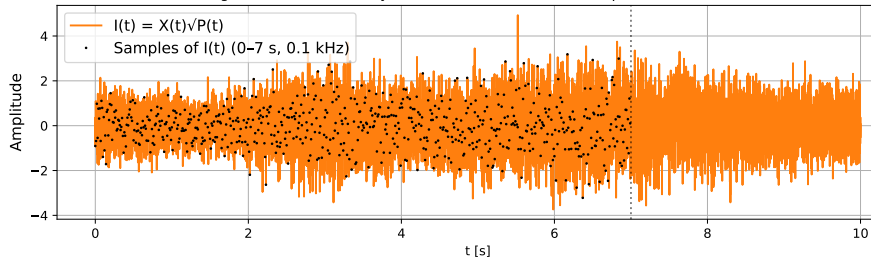
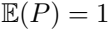


Fig 1(b): Non-stationary Gaussian noise $I(t)$ with sampled observations













expanding world + new world







Q1 2019









$$S_{\text{RX}}(t) = \frac{A}{l(d_0)} \sqrt{\frac{K}{K+1}} e^{-2i\pi\tau_0(t)f_c} e^{-2i\pi\theta} + \sum_{j=1}^{100} \frac{A}{l(d_j) \sqrt{100} \sqrt{K+1}} e^{-2i\pi\tau_j(t)f_c} e^{-2i\pi\theta},$$











2020 = 2020







QWERTY

$$v = \sqrt{\frac{GM}{h + R_{\oplus}}},$$

WELCOME TO THE









20

—

12

.

100



$$T_c = \frac{1}{8D_g} \approx 10^{-3}$$











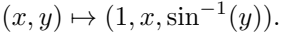






A row of 12 pixelated, grayscale characters. From left to right, they are: 'A', 'E', 'I', 'L', 'P', 'R', 'X', 'I', 'L', 'I', 'I', 'I'. The characters are rendered in a low-resolution, pixelated style with varying shades of gray and black, giving them a retro, digital appearance.











$$1 = e^{-\frac{\pi^2}{2\sqrt{P}}}$$





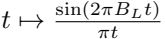
$$] = 1 - \left(1 - e^{-\frac{\pi^2}{2\sqrt{P/N}}} \right)^N,$$





192







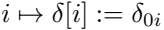
$$t \rightarrow \text{order} \rightarrow \frac{\sin(2\pi B t)}{\pi t}$$





$$\mathbb{Z} \ni i \mapsto \frac{\sin\left(2\pi \frac{B_L}{f_c} i\right)}{\pi i},$$

$$i \mapsto \delta[i] - \frac{\sin(2\pi \frac{B_H}{f_c} i)}{\pi i},$$



গণপ্রজাতন্ত্রী বাংলাদেশ

WAVELENGTH



WINNERS

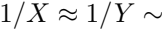
$$\mu = \exp\left(\mu_{\text{LIN}} + \frac{\sigma_{\text{LIN}}^2}{2}\right)$$

$$O(2\pi) \cong SO(2) \cong \mathbb{R}/2\pi\mathbb{Z}$$



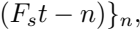






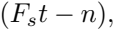
WORLD OF





1991-10-1

$$S(t) = \sum_{n=0}^{N-1} x[n]$$













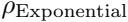


$$H = \begin{pmatrix} -a & -0 & -1 & 0 & 0 & 0 & ar & 0 & r \\ 0 & 0 & 0 & -a & -0 & -1 & 0 & 0 & 0 \\ -c & -b^2/a & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -c & -b^2/a & -1 & cr & rb^2/a & r \\ a & 0 & -1 & 0 & 0 & 0 & ar & 0 & -r \\ 0 & 0 & 0 & a & 0 & -1 & 0 & 0 & 0 \\ -c & b^2/a & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -c & b^2/a & -1 & -rc & rb^2/ar & -r \end{pmatrix}$$

$$\lim_{x \rightarrow \infty} - \frac{\log \mathbb{P}(X \geq x)}{x},$$







$$= \lim_{x \rightarrow \infty} - \frac{\log \mathbb{P}(X > x)}{x} = \lim_{x \rightarrow \infty} - \frac{\log(e^{-x/\theta})}{x}$$

$$= \lim_{x \rightarrow \infty} \frac{-x/\theta}{x} = 1/\theta.$$





$$= \lim_{x \rightarrow \infty} - \frac{\log \mathbb{P}(X > x)}{x} = \lim_{x \rightarrow \infty} - \frac{\log(1 - \gamma(k, x/\theta)/\Gamma(k))}{x}$$

$$= \lim_{x \rightarrow \infty} - \frac{\log(\Gamma(k, x/\theta)/\Gamma(k))}{x}$$

$$\begin{aligned}
 &\stackrel{(a)}{=} \lim_{x \rightarrow \infty} - \frac{\log(\Gamma(k, x/\theta))}{x} = \lim_{x \rightarrow \infty} - \frac{\log(x^{k-1} e^{-x/\theta})}{x}
 \end{aligned}$$

$$= \lim_{x \rightarrow \infty} - \frac{\log(e^{-x/\theta})}{x} = 1/\theta.$$



$$\lim_{x \rightarrow \infty} \frac{\Gamma(s, x)}{x^{s-1} e^{-x}} = 1.$$







$$\begin{aligned}
&= \lim_{x \rightarrow \infty} -\frac{\log \mathbb{P}(X > x)}{x} = -\lim_{x \rightarrow \infty} \frac{\log \operatorname{erfc}\left(\frac{x-\mu}{\sigma\sqrt{2}}\right)}{x} \stackrel{(b)}{\geq} -\lim_{x \rightarrow \infty} \frac{\log\left(\sqrt{\frac{e}{2\pi}} e^{-2\left(\frac{x-\mu}{\sigma\sqrt{2}}\right)^2}\right)}{x}
\end{aligned}$$

$$\geq -\lim_{x \rightarrow \infty} \frac{\log \left(e^{-2 \left(\frac{x-\mu}{\sigma \sqrt{2}} \right)^2} \right)}{x} = -\lim_{x \rightarrow \infty} \frac{-2 \left(\frac{x-\mu}{\sigma \sqrt{2}} \right)^2}{x} = \frac{1}{\sigma^2} \lim_{x \rightarrow \infty} \frac{x^2 - 2x\mu + \mu^2}{x} = \infty,$$



$$\phi(x) \geq \sqrt{\frac{2e}{\pi}} \frac{\sqrt{\beta - 1}}{\beta} e^{-\beta x^2},$$















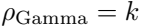










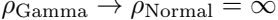














Q1 2024

2023年12月21日



$$\mathbb{E}[(S_1 + S_2)^2] = \int_0^1 (\cos(2\pi t) + \cos(2\pi t + \pi))^2 dt = \int_0^1 0 dt = 0,$$



1921 + 1921

$$= \int_0^1 \cos^2(2\pi t) dt + \int_0^1 \cos^2(2\pi t + \pi) dt$$

$$= \int_0^1 2 \cos^2(2\pi t) dt = \int_0^1 \cos(4\pi t) dt + 1 = 1.$$





$$E[S_1^2 + S_2^2] = E[S_1^2] + E[S_2^2] = E[S_1^2] + E[S_2^2]$$

1991 + 1991 + 1991

1992-93

$$\mathbb{E}[S_1 S_2] = \int_0^1 \cos(2\pi t) \cos(2\pi t + \pi) dt = \int_0^1 -\cos^2(2\pi t) dt = -\frac{1}{2} \int_0^1 \cos(4\pi t) dt - \frac{1}{2} = -\frac{1}{2},$$

Q1

2

Q2

E

Q2

2024 + 2025

2023-04-20

$$\mathbb{E} \phi_1 \phi_2 [\mathbb{E} [S_1^2 + S_2^2]] = \mathbb{E} \phi_1 [\mathbb{E} [S_1^2]] + \mathbb{E} \phi_2 [\mathbb{E} [S_2^2]] + \mathbb{E} \phi_1 \phi_2 [\mathbb{E} [S_1 S_2]]$$

$$= \frac{1}{2\pi} \int_0^{2\pi} \int_0^1 (\cos(2\pi t + \phi_1))^2 dt d\phi_1 + \frac{1}{2\pi} \int_0^{2\pi} \int_0^1 (\cos(2\pi t + \phi_2))^2 dt d\phi_2$$

$$+ \frac{1}{2\pi} \int_0^{2\pi} \int_0^{2\pi} \int_0^1 \cos(2\pi t + \phi_1) \cos(2\pi t + \phi_2) dt d\phi_1 d\phi_2$$

$$= \frac{1}{2\pi} \int_0^{2\pi} \int_0^1 (\cos(2\pi t + \phi_1))^2 dt d\phi_1 + \frac{1}{2\pi} \int_0^{2\pi} \int_0^1 (\cos(2\pi t + \phi_2))^2 dt d\phi_2 = 1/2 + 1/2 = 1.$$

1992



212122222





$${}_3F_2(1, 1, 1 + b; 2, 2; x) = \sum_{n=0}^{\infty} \frac{(1)_n (1)_n (1 + b)_n}{(2)_n (2)_n} \frac{x^n}{n!}$$

$$= \sum_{n=0}^{\infty} \frac{(1+b)_n}{(n+1)^2 n!} x^n = \frac{1}{b!} \sum_{n=0}^{\infty} \frac{(n+1)_b}{(n+1)^2} x^n$$

$$\underline{\underline{(a)}} \quad \frac{1}{b!} \sum_{n=0}^{\infty} \frac{\sum_{k=1}^b \begin{bmatrix} b \\ k \end{bmatrix} (n+1)^k}{(n+1)^2} x^n$$

$$\begin{aligned}
 &= \frac{1}{b!} \sum_{k=1}^b \left[\begin{matrix} b \\ k \end{matrix} \right] \sum_{n=0}^{\infty} \frac{x^n}{(n+1)^{2-k}} \stackrel{(b)}{=} \frac{1}{b!} \sum_{k=1}^b \left[\begin{matrix} b \\ k \end{matrix} \right] \frac{\operatorname{Li}_{2-k}(x)}{x},
 \end{aligned}$$



